PhonoSec: Speech Emotion Recognition on Live Calls for Cyber Crime Police

Exploring innovations that will shape our world tomorrow.

Start

CHALLENGES & CONCERNS

CHALLENGES:

- Achieving real-time processing with minimal delay (≤ 1–2 seconds)
- Ensuring high accuracy across different accents, languages & call qualities
- Handling background noise and multiple speakers in live conversations
- Designing robust feature extraction and optimized ML models for efficiency

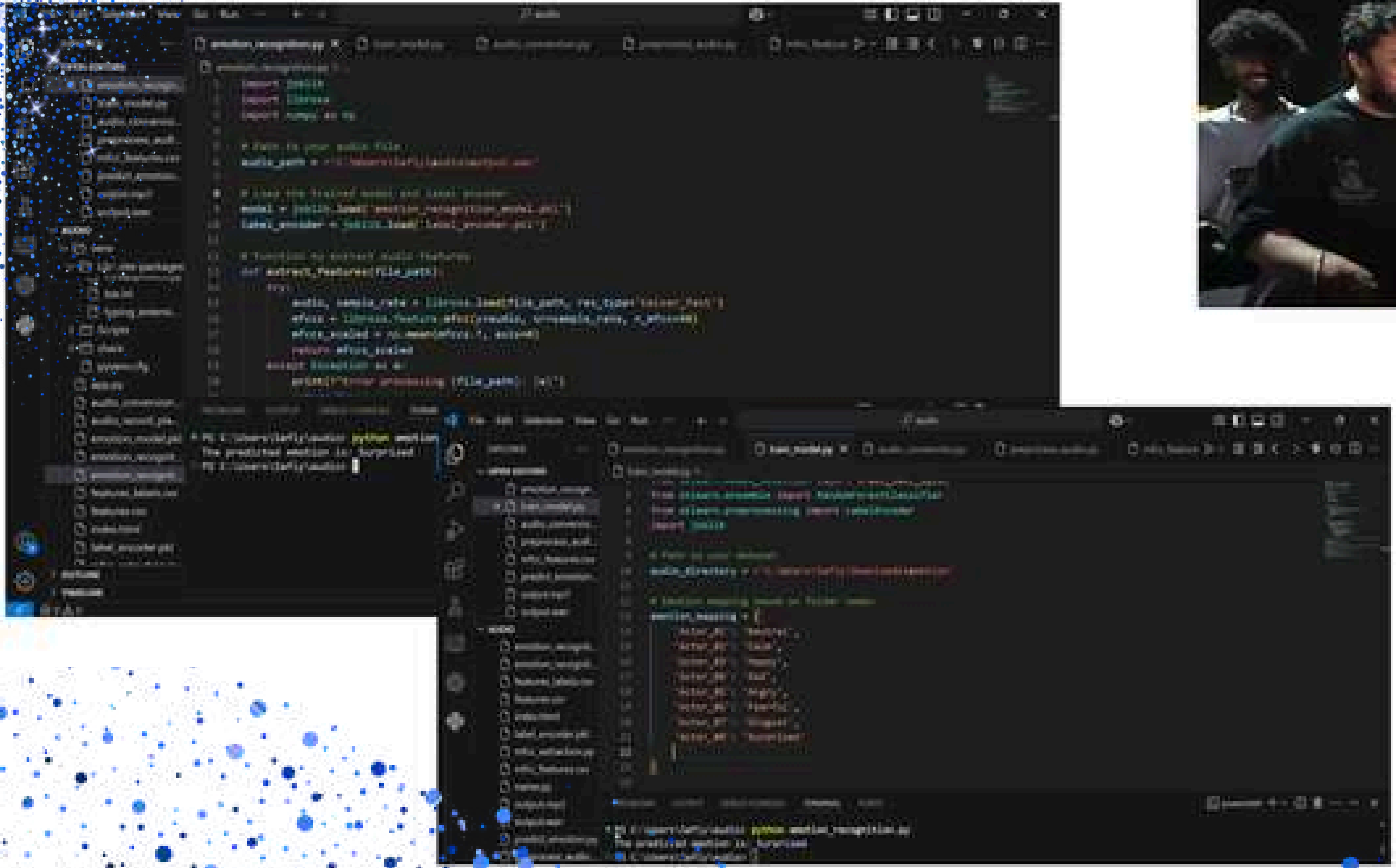
CONCERNS:

- Maintaining data privacy & legal compliance when analyzing sensitive live calls
- Preventing AI bias to ensure fair detection across all demographics
- Seamless integration into existing cybercrime police workflows
- Building trust, usability & ethical responsibility into the system

Our Solution: **PhonoSec**

PhonoSec is an AI-powered Speech Emotion Recognition system designed specifically for cybercrime police. It analyzes live call streams to detect emotions such as fear, anger, stress, and neutrality, helping officers identify distressed or high-risk callers in real time. By combining advanced voice feature extraction (MFCC) with machine learning models trained on labeled audio datasets, PhonoSec delivers fast and accurate predictions across different accents, background noise, and call qualities. The solution integrates seamlessly into existing call management systems, ensuring officers receive real-time emotional alerts without disrupting their workflow. Built with a focus on privacy and ethical AI, PhonoSec empowers cybercrime police to prioritize urgent cases, respond faster, and enhance victim protection.

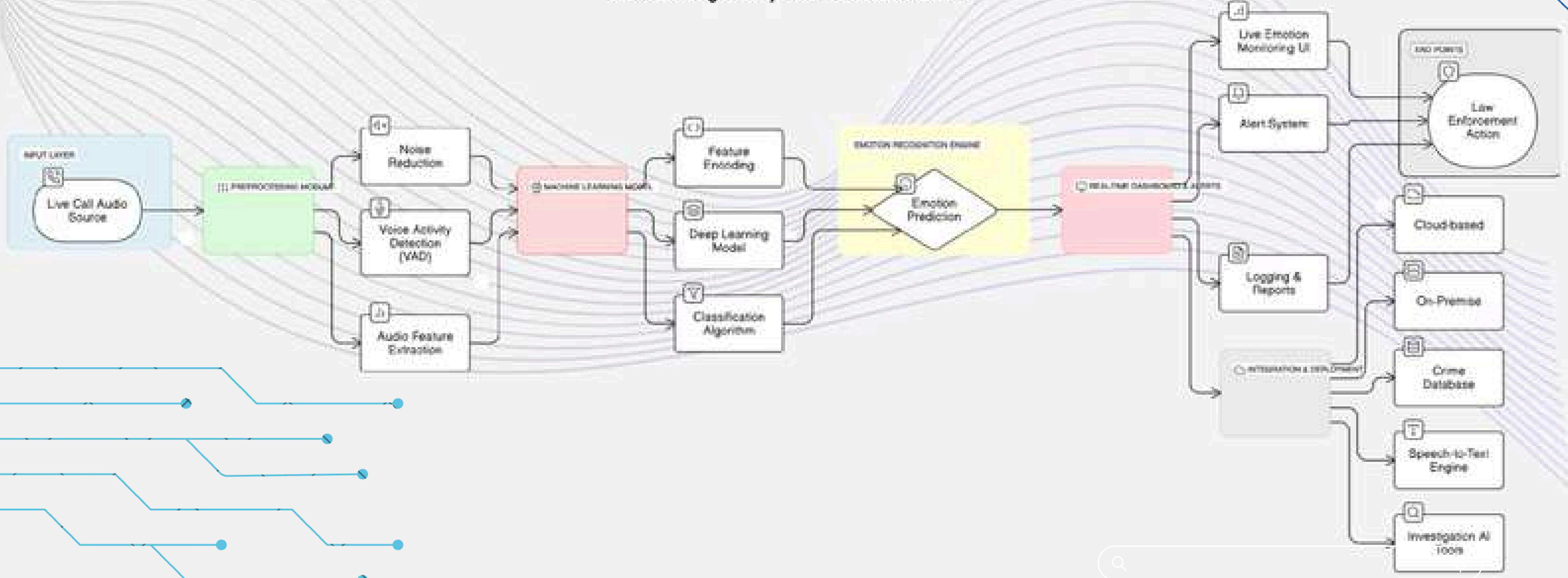
OUR EFFORTS & OUTPUT



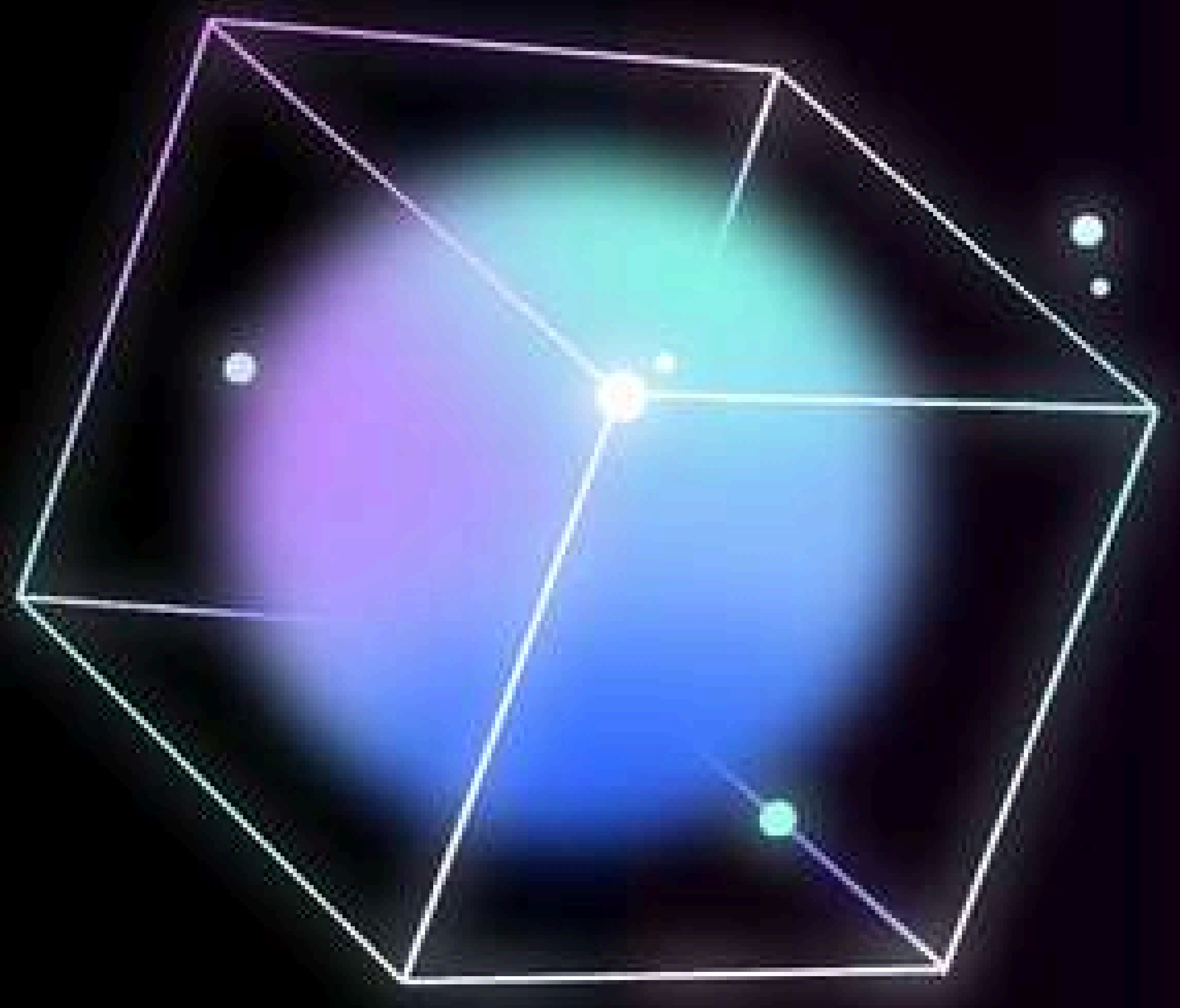
OUTPUT: SUPRISED

FLOW CHART

Emotion Recognition System for Law Enforcement



TECH STACK



Programming Language and Deployment	Programming Language: Python Deployment: Cloud-based or On-premise Integration
Libraries and Framework	Libraries: Librosa (Audio Processing), Scikit-learn, TensorFlow/Keras Framework: Flask/Django (For API & UI)
Machine Learning Model	Machine Learning Model: CNN/RNN

HOW IT WORKS

1. Audio Processing – Captures live call audio.
2. Feature Extraction – Extracts MFCC, pitch, and tone variations.
3. ML Model Prediction – Detects emotions (Angry, Happy, Sad, Fear, Neutral, etc.).
4. Real-Time Dashboard – Displays emotional trends and alerts for suspicious behavior.

KEY FEATURES

1. Real-Time Emotion Detection: Instant analysis of live call audio.
2. Automated Alerts: Notifies officers about distress signals or suspicious calls.
3. Integration with Cyber Crime Units: Can be used alongside call tracking & forensic tools.
4. Multilingual Support: Works with multiple languages & accents.
5. Scalable & Secure: Ensures data privacy & secure communication.

ENHANCED PROTECTION

Competitive Advantage

- Existing Solutions: Emotion detection in offline recordings.
- Our Innovation: Live call analysis with real-time AI-based insights.

Use Case Scenarios

- Emergency Helpline Monitoring: Detects genuine distress calls.
- Fraudulent Call Detection: Identifies scam calls based on emotion patterns.
- Threat Intelligence: Helps in analyzing caller intent during cyber investigations.

Future Scope

- Integration with law enforcement call tracking systems.
- AI-powered speech-to-text for additional forensic analysis.
- Deep learning models for detecting deception in speech.

Benefits:

- Faster response to cyber threats.
- Data-driven investigation approach.
- Reduces human bias in decision-making.

Project Timeline

1. Data Preparation & Preprocessing (2-3 Weeks)
2. Model Development & Training (4-6 Weeks)
3. API & UI Development (3-5 Weeks)
4. Deployment & Optimization (2-4 Weeks)
5. Testing & Finalization (2-3 Weeks)

Estimated Total Time: 3 to 4 Months

BUDGET SUMMARY

TOTAL PRODUCT BUDGET: ₹ 14,00,000

Category	Details	Estimated Cost (₹)
Model development & AI training	Advanced speech emotion models, feature engineering, accuracy tuning	₹ 3,50,000
Infrastructure & cloud	Servers / GPU instances, data storage, maintenance	₹ 2,50,000
Dataset acquisition & labeling	Purchase & cleaning of multilingual, diverse emotional speech datasets	₹ 2,00,000
Integration & deployment	API development, live call system integration	₹ 2,00,000
UI/UX & dashboard	Officer-friendly real-time emotion dashboard, visual alerts	₹ 1,00,000
Testing & compliance	Privacy, ethical AI audits, legal compliance (IT Act, GDPR)	₹ 1,00,000
Marketing & pilot rollout	Awareness workshops, training sessions with cybercrime units	₹ 50,000
Contingency (~10%)	Unforeseen technical / operational expenses	₹ 1,50,000

FUND UTILIZATION PLAN

We request ₹ 15,00,000 to cover the full project scope, pilot deployment, and first year support.

Use	%	Amount (₹)
AI development & data processing	30%	₹ 4,50,000
Infrastructure & cloud setup	20%	₹ 3,00,000
Integration & dashboard design	20%	₹ 3,00,000
Legal compliance & security	10%	₹ 1,50,000
Pilot rollout & marketing	10%	₹ 1,50,000
Contingency	10%	₹ 1,50,000

TOTAL: ₹ 15,00,000

Cyber Protect Data Threat Security Attack Firewall Malware

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